

Project Design Phase Solution Architecture

Date	02 October 2026
Team ID	SWTID-2026-9223
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

The proposed solution architecture connects the Standard Laptop Service Catalog, approval process, Flow Designer automation, Catalog Task creation, and Hardware assignment. It removes manual task creation and assignment after an approved laptop request.

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram and Data Flow:

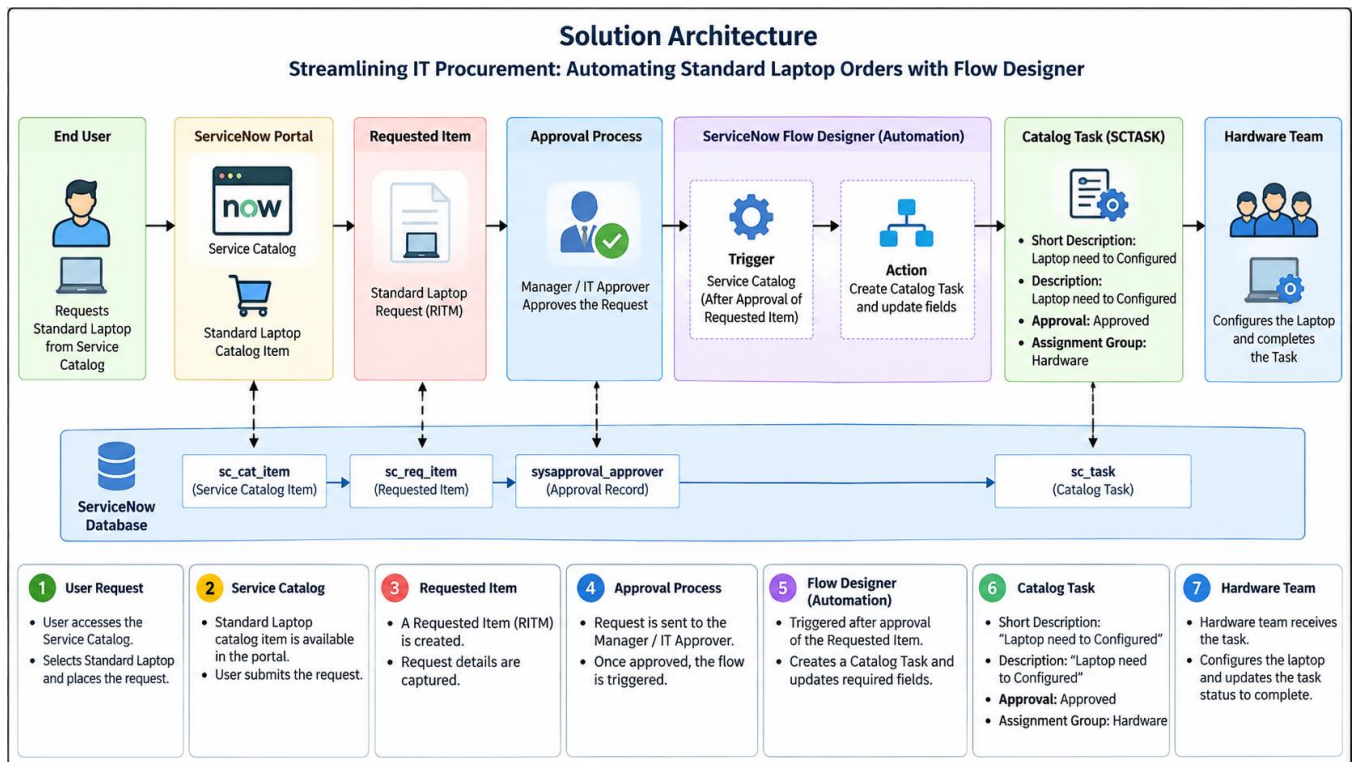


Figure 1: Solution architecture and data flow for automated Standard Laptop procurement and Hardware configuration using ServiceNow Flow Designer.

Architecture Components

Service Catalog: Provides the Standard Laptop catalog item through which the user submits the hardware request.

Requested Item: Stores the submitted Standard Laptop request and is used as the input record for the automated Catalog Task.

Approval: Ensures that the laptop request is approved before the configuration task is created.

Flow Designer: Executes the automation after the Service Catalog process reaches the required condition. The flow is configured to run as the System User.

Create Catalog Task: Creates a Catalog Task from the Requested Item Record for the laptop configuration activity.

Task Details: Sets the Short Description and Description to 'Laptop need to Configured' and the Approval value to 'Approved'.

Hardware Assignment Group: The generated Catalog Task is automatically assigned to Hardware so the responsible team can configure the laptop.

End-to-End Data Flow

The requester selects Standard Laptop from the Service Catalog and submits the order. A Requested Item is generated and proceeds through approval. After approval, Flow Designer creates a Catalog Task using the Requested Item, populates the required configuration details, and assigns the task to the Hardware group. The Hardware team then performs the laptop configuration.